Foundations of Stochastic gradient descent

[Stochastic gradient descent]

$$\theta_{t+1} = \theta_t - \eta \nabla_\theta L(\theta_t)$$

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w new := arg ■ min w { Q i (w) + 1 2 η ■ w − w old ■ 2 } . {\displaystyle w^{\text{new}}:=\arg \min _{w}\left\{Q_{i}(w)+{\frac {1}{2\eta }}\left\|w-w^{\text{old}}\right\|^{2}\right\}.}

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This equation is implicit since w new {\displaystyle w^{\text{new}}} appears on both sides of the equation. It is a stochastic form of the proximal gradient method since the update can also be written as:

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where ε {\displaystyle \varepsilon } is a small scalar (e.g. 10 − 8 {\displaystyle 10^{-8}}) used to prevent division by 0, and β 1 {\displaystyle \beta _{1}} (e.g. 0.9) and β 2 {\displaystyle \beta _{2}} (e.g. 0.999) are the forgetting factors for gradients and second moments of gradients, respectively. Squaring and square-rooting is done element-wise. As the exponential moving averages of the gradient m w (t) {\displaystyle m_{w}^{(t)}} and the squared gradient v w (t) {\displaystyle v_{w}^{(t)}} are initialized with a vector of 0's, there would be a bias towards zero in the first training iterations. A factor 1 1 − β 1 / 2 t {\displaystyle {\tfrac {1}{1-\beta _{1/2}^{t}}}} is introduced to compensate this bias and get better estimates m ^ w (t) {\displaystyle {\hat {m}}_{w}^{(t)}} and v ^ w (t) {\displaystyle {\hat {v}}_{w}^{(t)}} . The initial proof establishing the convergence of Adam was incomplete, and subsequent analysis has revealed that Adam does not converge for all convex objectives. Despite this, Adam continues to be used due to its strong performance in practice.

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max k = 0 , … , ■ T / η ■ | E [g (w k)] − g (W k η) | ≤ C η , {\displaystyle \max _{k=0,\dots ,\lfloor T/\eta \rfloor }\left|\mathbb {E} [g(w_{k})]-g(W_{k\eta })\right|\leq C\eta ,}

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RMSProp has shown good adaptation of learning rate in different applications. RMSProp can be seen as a generalization of Rprop and is capable to work with mini-batches as well opposed to only full-batches.

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max k = 0 , … , ■ T / η ■ | E [g (w k)] − E [g (W k η)] | ≤ C η 2 . {\displaystyle \max _{k=0,\dots ,\lfloor T/\eta \rfloor }\left|\mathbb {E} [g(w_{k})]-\mathbb {E} [g(W_{k\eta })]\right|\leq C\eta ^{2}.}

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d W t = − ∇ (Q (W t) + 1 4 η | ∇ Q (W t) | 2) d t + η Σ (W t) 1 / 2 d B t , {\displaystyle dW_{t}=-\nabla \left(Q(W_{t})+{\tfrac {1}{4}}\eta |\nabla Q(W_{t})|^{2}\right)dt+{\sqrt {\eta }}\Sigma (W_{t})^{1/2}dB_{t},}

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w new = w old + η 1 + η ■ x i ■ 2 (y i − x i ′ w old) x i . {\displaystyle w^{\text{new}}=w^{\text{old}}+{\frac {\eta }{1+\eta \left\|x_{i}\right\|^{2}}}\left(y_{i}-x_{i}'w^{\text{old}}\right)x_{i}.}

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[w 1 w 2] ← [w 1 w 2] − η [∂ ∂ w 1 (w 1 + w 2 x i − y i) 2 ∂ ∂ w 2 (w 1 + w 2 x i − y i) 2] = [w 1 w 2] − η [2 (w 1 + w 2 x i − y i) 2 x i (w 1 + w 2 x i − y i)] . {\displaystyle {\begin{bmatrix}w_{1}\\w_{2}\end{bmatrix}}\leftarrow {\begin{bmatrix}w_{1}\\w_{2}\end{bmatrix}}-\eta {\begin{bmatrix}{\frac {\partial }{\partial w_{1}}}(w_{1}+w_{2}x_{i}-y_{i})^{2}\\{\frac {\partial }{\partial w_{2}}}(w_{1}+w_{2}x_{i}-y_{i})^{2}\end{bmatrix}}={\begin{bmatrix}w_{1}\\w_{2}\end{bmatrix}}-\eta {\begin{bmatrix}2(w_{1}+w_{2}x_{i}-y_{i})\\2x_{i}(w_{1}+w_{2}x_{i}-y_{i})\end{bmatrix}}.} Note that in each iteration or update step, the gradient is only evaluated at a single x i {\displaystyle x_{i}} . This is the key difference between stochastic gradient descent and batched gradient descent. In general, given a linear regression y ^ = ∑ k ∈ 1 : m w k x k {\displaystyle {\hat {y}}=\sum _{k\in 1:m}w_{k}x_{k}} problem, stochastic gradient descent behaves differently when m < n {\displaystyle m<n} (underparameterized) and m ≥ n {\displaystyle m\geq n} (overparameterized). In the overparameterized case, stochastic gradient descent converges to arg ■ min w : w T x k = y k ∀ k ∈ 1 : n ■ w − w 0 ■ {\displaystyle \arg \min _{w:w^{T}x_{k}=y_{k}\forall k\in 1:n}\|w-w_{0}\|} . That is, SGD converges to the interpolation solution with minimum distance from the starting w 0 {\displaystyle w_{0}} . This is true even when the learning rate remains constant. In the underparameterized case, SGD does not converge if learning rate remains constant.

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Sign-based stochastic gradient descent Even though sign-based optimization goes back to the aforementioned Rprop, in 2018 researchers tried to simplify Adam by removing the magnitude of the stochastic gradient from being taken into account and only considering its sign. This results in a significantly lower communication cost of transferring gradients from workers to the parameter server. In this sense, it serves to better compress the gradient information, while having comparable convergence to standard SGD.

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subject to additional stochastic noise. This approximation is only valid on a finite time-horizon in the following sense: assume that all the coefficients Q i {\textstyle Q_{i}} are sufficiently smooth. Let T > 0 {\textstyle T>0} and g : R d → R {\textstyle g:\mathbb {R} ^{d}\to \mathbb {R} } be a sufficiently smooth test function. Then, there exists a constant C > 0 {\textstyle C>0} such that for all η > 0 {\textstyle \eta >0}

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Second-order methods A stochastic analogue of the standard (deterministic) Newton–Raphson algorithm (a "second-order" method) provides an asymptotically optimal or near-optimal form of iterative optimization in the setting of stochastic approximation. A method that uses direct measurements of the Hessian matrices of the summands in the empirical risk function was developed by Byrd, Hansen, Nocedal, and Singer. However, directly determining the required Hessian matrices for optimization may not be possible in practice. Practical and theoretically sound methods for second-order versions of SGD that do not require direct Hessian information are given by Spall and others. (A less efficient method based on finite differences, instead of simultaneous perturbations, is given by Ruppert.) Another approach to the approximation Hessian matrix is replacing it with the Fisher information matrix, which transforms usual gradient to natural. These methods not requiring direct Hessian information are based on either values of the summands in the above empirical risk function or values of the gradients of the summands (i.e., the SGD inputs). In particular, second-order optimality is asymptotically achievable without direct calculation of the Hessian matrices of the summands in the empirical risk function. When the objective is a nonlinear least-squares loss